

Oral Cavity in Diseases



Regressive Alterations & Resorption of Teeth

Lecture Learning Outcomes

By the end of this session students will be able to:

1. Define the terminologies related to **Regressive Alterations & Resorption of Teeth.**
2. Explain the Clinico-Pathological Features and Management of **Regressive Alterations & Resorption of Teeth..**

Attrition

Attrition is the wearing away of tooth structure (physiological or pathological) due to tooth-to-tooth contact, without involvement of external agents.

Etiopathogenesis

- Normal mastication (physiological)
- Bruxism (grinding/clenching)
- Malocclusion or improper bite
- Parafunctional habits

Clinico-Pathological Features

- Flattening of cusps and Incisal edges
- Reduced crown height
- Shiny wear facets
- Enamel Loss and Dentin exposure (yellowish appearance)
- Sensitivity of teeth
- Loss of vertical dimension of occlusion (VDO)
- **In Severe Cases:** Pulpal involvement, Tooth fracture



Diagnosis

- Patient history (grinding habits)
- Clinical examination (wear facets, symmetry)
- Radiographs (pulp size, Enamel and dentin thickness)



Principles of Management: Preventive and Restorative

- Patient education
- Stress management (for bruxism)
- Use of night guards/occlusal splints
- Crowns in severe wear
- Full mouth rehabilitation in advanced cases



Abrasion

Abrasion is the **pathological wearing away of tooth structure** due to **mechanical forces from external agents** (not tooth-to-tooth contact).

Etiopathogenesis

- Improper tooth brushing (horizontal/scrubbing technique)
- Use of hard-bristled toothbrush
- Abrasive toothpaste
- Habits (holding pins, nails, pipe smoking)

Clinico-Pathological Features

- More common on buccal surfaces of canines and premolars
- Shallow notches at cervical region of teeth (near CEJ) in initial stage and Deep V-shaped or wedge-shaped defects in advanced stages.
- Smooth, shiny surface usually painless initially followed by exposure of dentin leading to tooth sensitivity.



Diagnosis

- Clinical examination (location, shape of lesion)
- History of brushing habits
- Differentiate from other non-carious lesions



Principles of Management – Preventive & Restorative

- Educate proper brushing technique (vertical/circular)
- Use soft/Medium -bristled toothbrush
- Avoid abrasive toothpaste
- Eliminate harmful habits
- Desensitizing agents (for sensitivity)
- Restorations GIC & Composite



Erosion

Erosion is the progressive, irreversible loss of tooth structure caused by chemical dissolution by acids of non-bacterial origin, without involvement of mechanical forces.

Etiopathogenesis

- **Extrinsic acids:** Citrus fruits, carbonated drinks, sports drinks
- **Intrinsic acids:** Gastroesophageal Reflux Disease, chronic vomiting (e.g., eating disorders)
- **Environmental exposure:** Acidic industrial fumes, swimming in poorly buffered chlorinated pools
- **Medications:** Vitamin C tablets, aspirin etc.

Clinicopathological Features

- Smooth, shiny enamel surface with loss of surface detail
- Cupping of occlusal surfaces and shallow, rounded depressions
- Enamel thinning → dentin exposure
- Increased tooth sensitivity
- *In intrinsic erosion: palatal surfaces of maxillary teeth commonly affected.*



Diagnosis

- Detailed dietary and medical history suggesting acid exposure.
- Clinical appearance: rounded, non-angular lesions.
- Pattern of involvement (e.g., palatal vs labial surfaces)



Principles of Management

- Dietary counseling (reduce acidic intake).
- Medical referral for GERD or vomiting.
- Fluoride therapy and remineralizing agents.
- Desensitizing agents.
- Composite restorations or veneers.
- Full coverage crowns in severe tooth loss.



Internal Resorption

Internal resorption is a pathological, progressive loss of dentin from within the pulp chamber or root canal due to clastic activity of pulpal origin.

Etiopathogenesis

- Chronic pulpal inflammation (e.g., deep caries)
- Trauma to the tooth
- Pulpal infection
- Iatrogenic factors (heat during cavity preparation)
- Idiopathic (unknown cause in some cases)

Clinicopathological Features

- Usually asymptomatic in early stages
- Classic **“pink tooth”** appearance (pink spot) due to vascular granulation tissue
- Progressive loss of dentin from within the pulp chamber/canal
- Teeth are prone to fracture
- Perforation → communication with periodontal tissues.

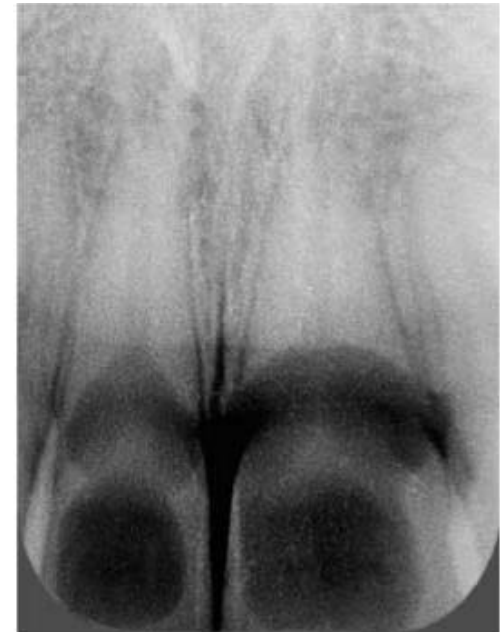


Diagnosis

- Often detected incidentally on radiographs
- well-defined, round/oval radiolucency continuous with pulp space
- Lesion remains centered on the canal in different angulated radiographs
- Vitality tests: tooth may remain vital in early stages
- Differentiate from external resorption (which shifts with angulation).

Principles of Management

- Endodontic treatment to remove vital/inflamed pulp
- Use of calcium hydroxide as an intracanal medicament.
- Obturation with thermoplasticized gutta-percha for irregular defects
- If perforation occurs → repair with materials like MTA
- Extraction in non-restorable or severely perforated cases



External Resorption

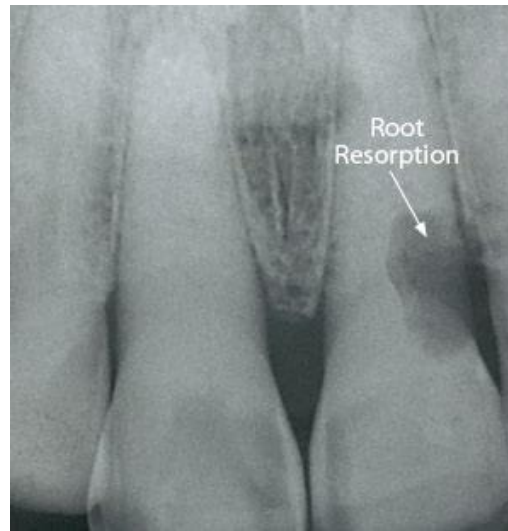
External resorption is the pathological loss of tooth structure (cementum and dentin) starting from the external surface of the tooth due to clastic activity.

Etiopathogenesis

- Trauma (luxation, avulsion injuries)
- Orthodontic tooth movement (excessive force)
- Periapical or periodontal inflammation
- Impacted teeth or pressure from tumors/cysts
- Reimplantation and transplantation of teeth
- Idiopathic causes

Clinicopathological Features

- Usually asymptomatic in early stages
- Irregular defects on root surface
- Progressive destruction of cementum/dentin
- Shortening of root (apical resorption)
- Mobility in advanced cases

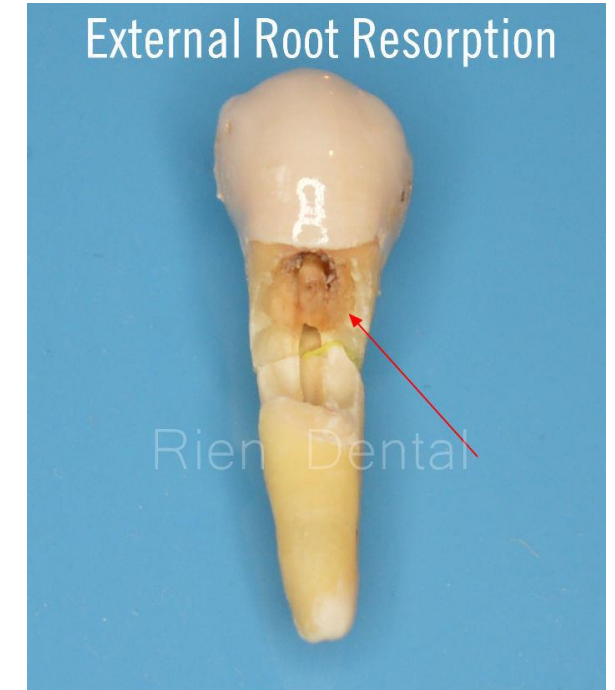


Diagnosis

- Detected on radiographs
- Irregular radiolucency
- Lesion shifts with change in angulation (parallax technique)
- Pulp vitality may remain normal (unless advanced case).
- CBCT may help in detailed assessment.

Principles of Management

- Monitoring in mild cases
- Eliminate causative factor (e.g., adjust orthodontic forces, treat infection)
- Endodontic treatment if pulp is involved
- Use of materials like MTA for repair
- Extraction in severe, non-restorable cases



- **KINDLY REFER**

Shafer's text book of oral pathology – 2020 Elsevier (9th edition).

Oral Pathology - Clinicopathologic Correlation Regezi, Sciubba & Jordan;; 2017 Elsevier (7th edition).